```c
#include <stdio.h>
#include <string.h>
#include "i2p.h"

#define TERMINATE 0
#define ADD_WORD 1
#define DISPLAY_WORDS 2
#define INCR_SORT 3
#define DECR_SORT 4
#define WORD_STATS 5
#define CHAR_STATS 6
#define SEARCH_WORD 7

#define MAX_WORDS 30
#define MAX_WORD_LEN 31

int menu(void);
int wordsLength;
int numOfWords;

void addWord(void);
void displayWords(void);
void incSort(void);
void decSort(void);
void wordStats(void);
void charStats(void);
void searchWord(void);
void getStringV1(char message[], char str[]);

void sortStringArrayInc(word, numOfWords, MAX_WORD_LEN);
void sortStringArrayDec(word, numOfWords, MAX_WORD_LEN);

char word[MAX_WORDS][MAX_WORD_LEN];

void main(){
int selectedOp;

printf("WordsHandling\n\n");
    selectedOp=menu();
        while(selectedOp!= TERMINATE){
            switch(selectedOp){
                case ADD_WORD :
                    getNumOfWords("Enter number of words: ");
                        addWord();
                    break;
                case DISPLAY_WORDS :
                    displayWords();
                    break;
                case INCR_SORT :
                    incSort();
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                    break;
                case DECR_SORT :
                    decSort();
                    break;
                case WORD_STATS :
                    wordStats();
                    break;
                case CHAR_STATS :
                    charStats();
                    break;
                case SEARCH_WORD :
                    searchWord();
                    break;
                default :
                    printf("Operation is not supported\n");
                    break;
                }
    selectedOp=menu();
}
printf("WordsHandling terminated\n");
return;
}

int menu(void){
int choice;

printf("\n\n-----------MENU--------------\n");
printf("0 - TERMINATE\n");
printf("1 - ADD WORD\n");
printf("2 - DISPLAY WORDS\n");
printf("3 - INCREMENTAL SORT\n");
printf("4 - DECREMENTAL SORT\n");
printf("5 - WORD STATISTICS\n");
printf("6 - CHARACTER STATISTICS\n");
printf("7 - SEARCH WORD \n----------------------------\n");
printf("Select operation:");
scanf("%d",&choice);
return (choice);
}

void addWord(void){
    int i;
    for(i=0;i<numOfWords;i++){
        printf("Add word: \n");
        scanf("%s", word[i]);
    }
}

void displayWords(void){
    printf("Display words: \n");
```

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101:     int j;
102:     for(j=0;j<numOfWords;j++)
103:         printf("%s\n", word[j]);
104: 
105: }
106: 
107: void incSort(void){
108:     int o=0;
109:     sortStringArrayInc(word, numOfWords, MAX_WORD_LEN);
110:     for(o=0;o<numOfWords;o++){
111:         printf("%s\n",word[o]);
112:     }
113: 
114: }
115: 
116: void decSort(void){
117:     sortStringArrayDec(word, numOfWords, MAX_WORD_LEN);
118:     int u=0;
119:     for(u=0;u<numOfWords;u++){
120:         printf("%s\n", word[u]);
121:     }
122: 
123: }
124: 
125: void wordStats(void){
126:     int u, o, max, min, overallLength;
127:     float avg;
128:     overallLength=strlen(word[0]);
129:     max=strlen(word[0]);
130:     min=strlen(word[0]);
131:     for(u=1;u<numOfWords;u++){
132:         if(strlen(word[u])>max){
133:             max=strlen(word[u]);
134:             }
135:         if(strlen(word[u])<min){
136:             min=strlen(word[u]);
137:             }
138:     }
139:     for(o=1;o<numOfWords;o++){
140:         overallLength+=strlen(word[o]);
141:     }
142:     avg=(float)overallLength/numOfWords;
143:     printf("Max word length = %d \nMin word length = %d \nAverage word length = %f",max,m
144:  }
145: 
146:  void charstats(void) {
147:     char schar;
148:     int totalCount = 0;
149:     int maxCount = 0;
150:     int minCount = 0;
```

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151:     int wordCounts[30];
152:     char maxWordchar[maxwordlen];
153:     char minWordchar[maxwordlen];
154:     float averageCount = 0;
155:
156:
157:     printf("Enter character to search: ");
158:     scanf("%c", &schar);
159:
160:
161:
162:     minCount = strlen(word[0]);
163:
164:
165:
166:     for (i = 0; i < numOfWords; i++) {
167:         int currentCount = 0;
168:
169:
170:         int j;
171:         for (j = 0; j < strlen(word[i]); j++) {
172:             if (word[i][j] == schar) {
173:                 currentCount++;
174:             }
175:         }
176:
177:
178:         wordCounts[i] = currentCount;
179:         totalCount += currentCount;
180:
181:         if (currentCount > maxCount) {
182:             maxCount = currentCount;
183:             strcpy(maxWordchar, word[i]);
184:         }
185:         if (currentCount < minCount) {
186:             minCount = currentCount;
187:             strcpy(minWordchar, word[i]);
188:         }
189:     }
190:
191:     if (numOfWords > 0) {
192:         averageCount = (float)totalCount / numOfWords;
193:     }
194:
195:     printf("Total character results'%c':\n", schar);
196:     printf("Total character occurencies: %d\n", totalCount);
197:     printf("Average number of occurencies per word: %.2f\n", averageCount);
198:     if (numberofWords > 0) {
199:         printf("Max number of occurencies in one word: %d (Word: '%s')\n", maxCount, maxW
200:         printf("Min number of occurencies in one word: leksi: %d (Word: '%s')\n", minCour
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201:     } else {
202:         printf("No words have been added to the list.\n");
203:     }
204: }
205:
206:  void searchWord(void) {
207:     char sword[MAX_WORD_LEN];
208:     int foundcount = 0;
209:     int theseis[30];
210:
211:     getStringV1("Enter word to search: ", sword);
212:     int i;
213:     for (i = 0; i < numOfWords; i++) {
214:         if (strcmp(word[i], sword) == 0) {
215:             theseis[foundcount] = i + 1;
216:             foundcount++;
217:         }
218:     }
219:
220:     if (foundcount > 0) {
221:         printf("Word '%s' found %d times on list.\n", sword, foundcount);
222:         printf("Found in positions below: ");
223:         for (i = 0; i < foundcount; i++){
224:             printf("%d ", theseis[i]);
225:         }
226:         printf("\n");
227:     } else {
228:         char choice;
229:         printf("Word '%s' not found.\n", sword);
230:         printf("Add word to list? (y/n)\n(y stands for yes and n stands for no,choose and
231:         scanf(" %c", &choice);
232:
233:         if (choice == 'y'){
234:             if (numOfWords < 30) {
235:                 strcpy(word[numOfWords], sword);
236:                 numOfWords++;
237:                 printf("Word '%s' added to list.\n", sword);
238:             } else {
239:                 printf("Word not added, list is full.\n");
240:             }
241:         } else {
242:             printf("Word not added.\n");
243:         }
244:     }
245: }
246:
247: int getNumOfWords(char message[]) {
248:     printf(message);
249:     scanf("%d", &numOfWords);
250:     return numOfWords;
```

```
251: }
252: int getInt(char message[]){
253:     printf(message);
254:     int num;
255:     scanf("%d", &num);
256:     return num;
257: }
```